

Problem – Solution Fit

Date	30 July 2026
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Project Name	Intergalactic Space Travel Request (ISTR) System
Maximum Marks	4 Marks

The Problem–Solution Fit canvas validates that the ISTR ServiceNow application directly addresses the real challenges faced by mission requesters, managers, and mission control staff during the space mission request and approval process.

Problem-Solution fit canvas 2.0

Purpose / Vision: To automate and simplify the intergalactic mission request and approval process by providing a secure, centralized, and workflow-driven ServiceNow application that improves efficiency, transparency, and mission management.

Define CS, fit into CC	1. CUSTOMER SEGMENT(S) CS <i>Who is your customer?</i> - Scientists - Astronauts - Research Officers - Mission Managers - Finance Officers - Mission Control Administrators - System Administrators - Space Exploration Organizations 	6. CUSTOMER CONSTRAINTS CC <i>What constraints prevent your customers from taking action or limit their choices of solutions? i.e. spending power, budget, no cash, network connection, available devices.</i> - Multiple approval levels - Manual documentation - Department dependency - Limited visibility of request status - Human errors - Approval delays - Lack of centralized records - Time constraints 	5. AVAILABLE SOLUTIONS AS <i>Which solutions are available to the customers when they face the problem or need to get the job done? What have they tried in the past? What pros & cons do these solutions have? i.e. pen and paper is an alternative to digital notetaking</i> Existing Alternatives: - Manual paper forms - Email-based approvals - Excel spreadsheets - Shared documents - Phone calls - Offline approval process Limitations: - Slow approvals - No workflow automation - No centralized tracking - Poor transparency - High manual effort - Difficult reporting 	Explore AS, differentiate
	2. JOBS-TO-BE-DONE / PROBLEMS J&P <i>Which jobs-to-be-done (or problems) do you address for your customers? There could be more than one, explore different sides.</i> - Submit intergalactic mission requests. - Obtain approvals from multiple departments. - Track mission request status. - Upload mission documents. - Receive approval notifications. - Manage mission records centrally. - Monitor completed and pending missions. - Generate mission reports and dashboards. 	9. PROBLEM ROOT CAUSE RC <i>What is the real reason that this problem exists? What is the back story behind the need to do this job? i.e. customers have to do it because of the change in regulations.</i> - No centralized mission management system. - Manual approval workflows. - Separate departmental processes. - Lack of automation. - Poor communication between stakeholders. - No real-time mission tracking. - Limited reporting capabilities. - Inefficient document management. 	7. BEHAVIOUR BE <i>What does your customer do to address the problem and get the job done? i.e. directly related: find the right solar panel installer, calculate usage and benefits; indirectly associated: customers spend free time on volunteering work (i.e. Greenpeace)</i> - Completes mission request forms manually. - Sends documents through email. - Contacts managers for approval updates. - Follows up with finance and mission control. - Maintains separate mission records. - Repeatedly checks request status. - Stores mission documents in multiple locations. 	
3. TRIGGERS TR <i>What triggers customers to act? i.e. seeing their neighbour installing solar panels, reading about a more efficient solution in the news.</i> - Need to plan a new space mission. - Mission launch schedule announced. - Research expedition approved. - Requirement to travel to another planet. - Existing manual approval process causes delays. - Need for faster request processing. 	10. YOUR SOLUTION SL <i>If you are working on an existing business, write down your current solution first, fill in the canvas, and check how much it fits reality. If you are working on a new business proposition, then keep it blank until you fill in the canvas and come up with a solution that fits within customer limitations, solves a problem and matches customer behaviour.</i> ISTR – Intergalactic Space Travel Request System The ISTR application is a ServiceNow-based solution that digitizes the complete mission request lifecycle. It enables users to submit mission requests through the Service Catalog, automates multi-level approvals using Flow Designer, validates data with Client Scripts and Business Rules, secures access through ACLs and Roles, sends automated email notifications, and provides real-time dashboards and reports. The solution improves transparency, reduces processing time, minimizes manual work, and delivers a centralized platform for efficient mission management. 	8. CHANNELS of BEHAVIOUR CH 8.1 ONLINE <i>What kind of actions do customers take online? Extract online channels from #7</i> - ServiceNow Service Portal - Service Catalog - Email Notifications - Dashboards - Reports - Internal Enterprise Network  8.2 OFFLINE <i>What kind of actions do customers take offline? Extract offline channels from #7 and use them for customer development.</i> - Face-to-face approvals - Paper documents - Department meetings - Printed mission records - Manual signatures - Phone communication 	Extract online & offline Cl of BE	
4. EMOTIONS: BEFORE / AFTER EM <i>How do customers feel when they face a problem or a job and afterwards? i.e. lost, insecure > confident, in control – use it in your communication strategy & design.</i> BEFORE - Frustrated - Uncertain - Confused - Time-consuming - Difficult to track AFTER - Confident - Organized - Informed - Efficient - Secure - Satisfied 	Identify strong TR & EM			



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1. Customer Segments (CS)

- Scientists
- Astronauts
- Mission Managers
- Finance Officers
- Mission Control Officers
- System Administrators

2. Jobs-to-be-Done / Problems (J&P)

- Submit mission requests
- Obtain multi-level approval
- Track request and mission status

3. Triggers (TR)

- Mission planning cycles
- Launch schedules
- Research expedition approval needs

4. Emotions (EM)

Before	After
Will my request even be approved?	Confident – automated workflow gives clear status
Frustrated by manual process	Relieved – no more manual follow-ups
Uncertain about approval timeline	Informed – real-time tracking

5. Available Solutions (AS)

- Paper forms
- Emails
- Excel spreadsheets

Limitations: no automation, no centralized tracking, poor visibility, slow multi-department coordination.

6. Customer Constraints (CC)

- Manual approvals across departments
- Department dependency for sign-off
- Poor visibility into request status

7. Behaviour (BE)

- Manual submissions via paper or email
- Repeated email follow-ups with approvers
- Manually tracking approval status across departments

8. Channels of Behaviour (CH)

Online	Offline
Service Portal	Face-to-face approvals
Service Catalog	Printed mission documents

Online	Offline
Email Notifications	Department meetings
Dashboard & Reports	Manual sign-offs

9. Problem Root Cause (RC)

- No centralized mission request system
- Manual, disconnected approval workflow
- Lack of automation across departments
- No real-time mission or approval tracking

10. Solution (SL)

ISTR – Intergalactic Space Travel Request System: a ServiceNow application built on a custom table (extending Task) that digitizes the full mission request lifecycle — Service Catalog intake, Flow Designer-driven multi-level approvals (Manager → Finance → Mission Control), Business Rule and Client Script validation, ACL-secured role-based access, automated email notifications, and a Performance Analytics dashboard with supporting reports.

Summary Table

Element	Details
Customer	Scientists, Astronauts, Mission Managers
Jobs	Mission Requests, Approval, Tracking
Available Solutions	Paper Forms, Emails, Excel
Customer Constraints	Manual approvals, department dependency, poor visibility
Behaviour	Manual submissions, email follow-ups
Triggers	Mission planning, launch schedules
Solution	ISTR ServiceNow Application